

# Toward End-to-End Intent-Based Networking for 6G: Architectures, Challenges, Recent Advances, and Future Directions

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**Abstract**—Intent-Based Networking (IBN) has emerged as a promising paradigm for simplifying network management by allowing operators and applications to specify high-level service objectives rather than low-level device configurations. Early IBN research was mainly developed in Software-Defined Networking (SDN), Network Function Virtualization (NFV), transport networks, core networks, and data-center environments, where programmability, virtualization, and relatively stable infrastructure models enabled intent translation, orchestration, and assurance. However, realizing IBN in end-to-end mobile networks is more challenging because the Radio Access Network (RAN) is highly dynamic, wireless-channel-dependent, mobility-sensitive, and governed by multiple control timescales. The emergence of Open RAN (O-RAN) changes this landscape by making the RAN programmable, disaggregated, data-driven, and controllable through non-real-time and near-real-time intelligent control loops. This survey reviews the evolution of IBN from SDN/NFV-enabled automation toward O-RAN-driven end-to-end intent-based networking for 5G-Advanced and 6G. We discuss architectural mechanisms, key challenges, recent advances in AI-driven and agentic IBN, and future research directions including Large Language Model (LLM)-based intent translation, contract-based O-RAN slicing, digital twin-assisted validation, and trustworthy closed-loop orchestration.

**Index Terms**—Intent-Based Networking, SDN, NFV, 6G, O-RAN, network slicing, LLM, agentic AI, closed-loop automation.

## I. INTRODUCTION

INTENT-BASED Networking (IBN) represents a shift in network management from device-level configuration to outcome-driven operation. Instead of requiring operators to manually specify how routers, switches, virtual functions, or radio parameters should be configured, IBN allows them to declare what service behavior should be achieved. The network management system then translates the declared intent into policies, configurations, orchestration decisions, and continuous assurance mechanisms [1]–[5].

The early development of IBN was closely tied to Software-Defined Networking (SDN), Network Function Virtualization (NFV), core networks, transport networks, and data-center infrastructures. These environments provided a natural starting point because they expose programmable control interfaces, virtualized network functions, and relatively structured resource models. In such domains, intent can be mapped to connectivity requirements, forwarding policies, service function chains, virtual network functions, resource allocation decisions, and verification rules. Early platforms and frameworks such as the ONOS Intent Framework, OpenDaylight NEMO, Network Intent Composition (NIC), and data-center-oriented IBN systems demonstrated how high-level requirements could be translated into lower-level network configurations [2], [6]–[9].

However, this SDN/NFV-centric view does not fully capture the requirements of next-generation mobile networks. In mobile systems, the Radio Access Network (RAN) is not merely an access segment but a major source of service variability. Radio channels fluctuate over time and space, users move across cells, traffic demand changes rapidly, and control decisions must be made across heterogeneous timescales ranging from fast radio control to long-term service policy optimization. As a result, realizing an end-to-end intent requires not only core, transport, and cloud orchestration, but also RAN-aware interpretation, monitoring, optimization, and assurance.

The emergence of Open RAN (O-RAN) provides a key architectural opportunity to extend IBN from SDN/NFV-enabled domains toward the RAN. By disaggregating RAN functions, exposing open interfaces, and introducing the non-real-time and near-real-time RAN Intelligent Controllers (RICs), O-RAN makes the RAN programmable and data-driven [10]–[12]. This programmability enables high-level intents to be de-

composed into RAN policies, rApp-level optimization logic, xApp-level control objectives, slicing constraints, mobility management rules, and radio resource optimization actions. Therefore, O-RAN can be viewed as a missing bridge between conventional IBN and end-to-end intent-based mobile network operation.

Recent advances in Artificial Intelligence (AI), especially Large Language Models (LLMs) and agentic AI, further accelerate this transition. LLMs can support natural-language intent interpretation, while agentic frameworks can decompose high-level objectives into domain-specific subtasks across RAN, transport, core, edge, and cloud domains. Recent works have investigated LLM-based RAN management, agentic AI-powered IBN for 6G, AI-driven self-configuration, and contract-based agentic intent frameworks for O-RAN slicing [13]–[16]. These studies suggest that IBN is evolving from SDN/NFV-based automation toward trustworthy, multi-domain, and O-RAN-enabled closed-loop intelligence.

The main contributions of this paper are summarized as follows:

- We review the evolution of IBN from SDN/NFV-enabled automation in core, transport, and data-center networks toward end-to-end intent-based networking for mobile systems.
- We explain why RAN environments introduce new challenges for IBN, including wireless dynamics, mobility, multi-timescale control, and service-aware radio resource management.
- We discuss how O-RAN programmability enables the extension of IBN to the RAN through non-real-time and near-real-time control loops, rApps, xApps, and open interfaces.
- We analyze recent advances in AI-driven, LLM-based, and agentic IBN, with emphasis on RAN management, self-configuration, and contract-based O-RAN slicing.
- We identify future research directions for end-to-end IBN, including digital twin-assisted validation, trustworthy agentic orchestration, hierarchical closed-loop control, and intent-aware network slicing.

## II. FROM SDN/NFV-BASED AUTOMATION TO END-TO-END IBN

### A. SDN/NFV as the Starting Point of IBN

SDN and NFV are the two major architectural shifts that enabled modern network automation. SDN separates the control plane from the data plane and allows a logically centralized controller to program forwarding behavior through southbound interfaces [17]. NFV decouples network functions from dedicated hardware

and allows functions such as firewalls, gateways, load balancers, and packet gateways to run on general-purpose computing platforms [18]. Together, SDN and NFV provide programmability, abstraction, and flexible service composition.

These properties made SDN/NFV-enabled environments a natural starting point for IBN. In a programmable transport or data-center network, high-level connectivity intents can be translated into paths, flow rules, access policies, or resource reservations. In an NFV environment, service intents can be translated into virtual network function placement, scaling, migration, and service function chaining decisions. Thus, SDN and NFV should be understood not as the final scope of IBN, but as the architectural foundation that made the first practical realization of IBN feasible.

However, SDN and NFV mainly expose how the network can be programmed. They do not inherently determine what configuration should be produced from a high-level objective. IBN adds a higher-level intelligence layer above SDN/NFV. This layer is responsible for understanding intent, deriving policies, resolving conflicts, selecting feasible actions, and verifying whether the operational state satisfies the intended outcome.

### B. Intent Lifecycle

A typical IBN lifecycle consists of five stages: intent expression, intent translation, orchestration, activation, and assurance. Intent expression captures the objective from a user, operator, application, or business system. The input may be a structured template, a domain-specific language, an API request, or natural language. Intent translation then converts the high-level objective into machine-understandable policies, constraints, and Key Performance Indicators (KPIs). For example, “high-quality immersive service” may be translated into bandwidth, delay, jitter, reliability, and packet loss requirements.

The orchestration stage decomposes the translated intent across multiple domains, such as transport, core, edge computing, cloud infrastructure, and RAN. Activation enforces the selected policies through SDN controllers, NFV orchestrators, RAN controllers, or network management systems. Finally, assurance continuously monitors the network and verifies whether the operational state satisfies the intended state. If a deviation is detected, the system may trigger re-optimization, reconfiguration, healing, or rollback.

This lifecycle shows why IBN is more than automation. Conventional automation can execute predefined actions, while IBN must continuously close the loop

between high-level objectives and actual network behavior. The next challenge is to extend this closed-loop lifecycle beyond relatively stable wired and virtualized infrastructures toward dynamic mobile networks.

### C. Need for End-to-End Intent Realization

In next-generation networks, service quality is determined by the interaction of multiple domains. A low-latency XR service, for example, may depend on RAN scheduling, mobility management, transport routing, edge computing placement, and core network function scaling. If each domain is optimized independently, one domain may improve its local KPI while degrading the end-to-end service objective.

End-to-end IBN therefore requires cross-domain intent decomposition and coordination. Velasco et al. proposed an end-to-end IBN framework in which RAN, transport, and cloud/edge domains are coordinated through service orchestration and ML-powered closed loops [19]. This direction is important because service degradation detected in one domain may require reconfiguration in another domain. For example, an increase in end-to-end delay observed at the application layer may be caused by RAN congestion, transport bottlenecks, or insufficient edge computing resources. Hence, intent assurance must be service-aware, multi-domain, and closed-loop.

## III. O-RAN-ENABLED INTENT-BASED NETWORKING

### A. Why RAN Makes IBN Difficult

Extending IBN to the RAN is significantly more difficult than applying it to data centers, transport networks, or virtualized core networks. First, the RAN is affected by time-varying wireless channels, interference, mobility, traffic burstiness, and spatially heterogeneous user distributions. Second, RAN control operates across multiple timescales, including long-term policy management, near-real-time control, and fast radio operations. Third, radio resources are tightly coupled with service-level objectives such as latency, reliability, energy efficiency, and Quality of Experience (QoE). Fourth, the impact of a control action is often uncertain because the wireless environment changes continuously.

These characteristics make RAN intent realization challenging. An intent such as “reduce energy consumption without degrading premium users” cannot be realized by a simple static policy. It may require dynamic cell activation, power control, mobility steering, slice-aware scheduling, and continuous QoE monitoring. Similarly, an intent such as “guarantee low-latency

industrial traffic” must be translated into radio resource policies, transport constraints, edge placement decisions, and slice isolation rules. Therefore, RAN-aware IBN requires both programmability and intelligence.

### B. O-RAN as a Programmable RAN Foundation

O-RAN provides the architectural foundation for bringing IBN into the RAN. The Service Management and Orchestration (SMO) framework and the Non-Real-Time RIC can host high-level policy generation, model training, long-term optimization, and intent interpretation. The Near-Real-Time RIC can enforce near-real-time control through xApps over a shorter control timescale. This creates a hierarchical control structure in which an operator-level intent can be decomposed into rApp-level policies, xApp-level control objectives, and E2-node configurations.

This O-RAN-based hierarchy changes the role of IBN. In earlier SDN/NFV environments, IBN mainly focused on connectivity, service chaining, and virtual resource orchestration. In O-RAN-enabled mobile networks, IBN must additionally handle radio resource allocation, mobility management, interference control, energy saving, slice assurance, and QoE-aware RAN optimization. As a result, O-RAN extends the applicability of IBN from core, transport, and data-center automation toward end-to-end programmable mobile networks.

### C. Toward End-to-End IBN for 6G

For 6G, IBN should not be limited to a single domain. Instead, it should operate as an end-to-end intent management framework across RAN, transport, core, edge, cloud, and possibly non-terrestrial network domains. In this architecture, high-level intents are first interpreted and decomposed into domain-specific sub-intents. The RAN sub-intent may be handled by the O-RAN control hierarchy, the transport sub-intent by SDN controllers, the core and edge sub-intents by NFV or cloud-native orchestrators, and the application-level sub-intent by service management systems.

The key architectural issue is coordination. Each domain may have its own local controller, telemetry abstraction, optimization objective, and control timescale. Therefore, end-to-end IBN requires a global intent layer that coordinates domain-specific controllers while preserving local autonomy. This suggests a hierarchical architecture: a global intent orchestrator interprets high-level service objectives and decomposes them into sub-intents, while domain-specific controllers realize and assure these sub-intents under local constraints.

#### IV. RECENT ADVANCES IN AI-DRIVEN AND AGENTIC IBN

##### A. LLM-Based Intent Translation for RAN Management

LLMs have strong potential for intent translation because they can understand natural language, extract objectives, identify constraints, and produce structured intermediate representations. Recent work on LLM-based IBN for RAN management demonstrates the feasibility of using LLMs to interpret high-level RAN management objectives and generate candidate configurations through structured reasoning and feedback [13]. This direction is important because RAN intents are often expressed in operational language, such as reducing energy consumption, improving coverage, or maintaining service quality for a target user group.

However, LLMs should not directly actuate the network. Natural-language outputs may be ambiguous, incomplete, or unsafe. Therefore, LLM-based IBN should include schema-constrained generation, domain-specific validation, policy checking, and human-in-the-loop supervision for high-risk actions. The LLM should act as an interpreter and planner, while the final policy must be validated by deterministic or formally defined mechanisms.

##### B. Agentic AI for Multi-Domain IBN

Agentic AI extends LLM-based IBN by allowing multiple specialized agents to reason over different network domains. For example, an orchestrator agent may decompose a high-level 6G service intent into RAN, transport, core, and edge subtasks. Domain-specific agents can then reason over local telemetry, constraints, and control actions. A verifier or auditor agent can check whether the proposed actions satisfy service-level and safety constraints.

Recent agentic AI-powered IBN research proposes such multi-agent reasoning for 6G networks [14]. This approach is attractive because future networks are inherently multi-domain. A single monolithic model cannot reliably understand all details of RAN control, core network orchestration, transport routing, cloud-native deployment, and non-terrestrial connectivity. Agentic decomposition allows each agent to focus on a domain while maintaining coordination through a higher-level orchestrator.

##### C. Self-Configuration and Proactive Assurance

Another recent direction is AI-driven self-configuration. Conventional IBN systems often focus on translating and deploying an intent, but next-generation IBN must also predict whether the intent will continue to be satisfied under future

network conditions. AI-driven self-configuration frameworks aim to combine intent translation, policy realization, conflict-aware activation, and proactive failure prediction [15].

This direction is particularly important for 6G services because reacting after service degradation may be too late. Mission-critical applications, industrial control, and immersive services require proactive assurance. Therefore, future IBN systems should predict possible intent violations before they occur and trigger preventive reconfiguration.

##### D. Contract-Based Agentic IBN for O-RAN Slicing

Contract-based agentic IBN has recently been proposed for O-RAN network slicing [16]. The key idea is to separate probabilistic intent extraction from deterministic policy execution. An LLM or agentic system may interpret user objectives and generate candidate slice configurations, but these candidates must be checked against formal RAN constraints, resource limits, and slice contracts before actuation.

This approach is highly relevant to O-RAN-enabled IBN. Network slicing requires resource isolation, SLA assurance, and fairness among tenants. If an AI agent generates an unsafe slice configuration, it may degrade other slices or violate isolation constraints. Contract-based validation provides a safety boundary for agentic decision-making. Therefore, contract-based IBN can serve as a bridge between flexible natural-language intent interpretation and trustworthy RAN control.

#### V. OPEN CHALLENGES AND FUTURE DIRECTIONS

##### A. Semantic Ambiguity and Intent Translation

One of the fundamental challenges in IBN is the semantic gap between user intent and network control. Users often express intents in high-level terms such as “low latency,” “high reliability,” “good coverage,” or “energy saving.” These terms are context-dependent and must be translated into measurable KPIs, constraints, and policies. In RAN environments, this translation becomes more difficult because the same intent may imply different radio configurations depending on mobility, traffic load, channel condition, and service type.

A promising direction is to combine LLM-based interpretation with domain ontologies, structured intent schemas, and formal validation. The LLM can extract objectives and constraints, while the schema and verifier ensure that the generated intent representation is complete, feasible, and safe.

### B. Multi-Domain Intent Decomposition

End-to-end IBN requires decomposing a single service intent into multiple domain-specific sub-intents. This is difficult because RAN, transport, core, edge, and cloud domains have different abstractions, control interfaces, and timescales. A low-latency service intent may require RAN prioritization, transport path optimization, edge function placement, and core network scaling. If these sub-intents are inconsistent, the end-to-end objective may fail.

Future research should investigate hierarchical decomposition methods that preserve the relationship between global service objectives and local domain actions. In particular, O-RAN-based RAN control should be coordinated with SDN/NFV-based transport and core orchestration to support end-to-end assurance.

### C. Conflict Resolution and Safe Control

Multiple intents may coexist in the same network. Some intents may compete for limited resources or impose contradictory constraints. For example, one intent may seek to minimize energy consumption, while another requires high coverage and low latency. In O-RAN slicing, different tenants may request overlapping radio resources while demanding strict isolation.

Conflict resolution should therefore be a first-class function in IBN. Future systems need mechanisms for detecting resource-level, policy-level, action-level, and impact-level conflicts. AI-based prediction can estimate the impact of candidate actions, while formal or contract-based methods can prevent unsafe actuation.

### D. Digital Twin-Assisted Intent Validation

Digital twin-assisted validation is a promising direction for trustworthy IBN. Before deploying a configuration to the physical network, the IBN system can test candidate actions in a network digital twin. This enables what-if analysis, policy validation, failure prediction, and rollback planning. For O-RAN, a RAN digital twin can estimate the impact of mobility steering, energy-saving policies, slice reconfiguration, or radio resource allocation before applying them to live cells.

For 6G, digital twin-assisted IBN is particularly important because networks will include terrestrial, aerial, satellite, edge, and cloud domains. Direct experimentation in live networks may be risky. A digital twin can provide a safe environment for evaluating whether an intent can be realized without causing SLA violations or cross-domain conflicts.

### E. Hierarchical Closed-Loop Control

IBN should not be implemented as a single global control loop. Different decisions operate at different timescales. Business-level intent may change over minutes or hours, network slice management may operate over seconds or minutes, near-real-time RAN control may operate over milliseconds to seconds, and lower-layer radio operations may operate even faster.

Future IBN architectures should explicitly support hierarchical closed-loop control. The global intent layer should handle service-level objectives and cross-domain decomposition. The O-RAN Non-RT RIC can support long-term RAN policy and model management, while the Near-RT RIC can perform near-real-time control through xApps. SDN and NFV controllers can manage transport, core, and cloud resources. This hierarchical structure is necessary for scalable and stable end-to-end IBN.

### F. Trustworthy Agentic Orchestration

Agentic AI introduces new opportunities and risks. Multi-agent systems can decompose complex intents and coordinate domain-specific reasoning, but they may also produce inconsistent or unsafe actions. Future agentic IBN systems should include explicit role separation among intent interpreter, domain expert, planner, verifier, executor, and auditor.

A trustworthy agentic IBN architecture should satisfy several principles. First, natural-language interpretation should be separated from policy enforcement. Second, all generated actions should be validated against constraints. Third, the system should maintain logs and explanations for accountability. Fourth, rollback mechanisms should be prepared before actuation. Fifth, human operators should remain involved for high-risk or large-scale actions.

## VI. CONCLUSION

IBN has evolved from a management abstraction for SDN/NFV-enabled networks into a broader paradigm for autonomous end-to-end network operation. Its early development was rooted in core networks, transport networks, and data-center infrastructures, where programmability and virtualization made intent realization feasible. However, the next stage of IBN lies in extending this paradigm toward mobile networks, especially the RAN.

O-RAN is a key enabler of this transition. By exposing programmable RAN control loops through the SMO, Non-RT RIC, Near-RT RIC, rApps, and xApps, O-RAN provides the missing architectural bridge between high-level service intent and radio-level control.

This enables IBN to move beyond SDN/NFV-based automation toward end-to-end intent-based networking across RAN, transport, core, edge, and cloud domains.

Recent advances in LLMs and agentic AI further strengthen this direction by enabling natural-language intent interpretation, multi-domain decomposition, and closed-loop policy reasoning. Nevertheless, trustworthy deployment remains challenging. Future IBN systems should combine AI-driven reasoning with contract-based constraints, digital twin-assisted validation, formal verification, and hierarchical control. In this direction, IBN can become a central management paradigm for 6G networks, where service objectives are continuously translated, enforced, monitored, and optimized across the entire network.

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